

EN401 — Energy Systems Modelling

Repository Architecture, System Design & Modelling Methodology Guide

COURSE: EN401 Energy Systems Modelling

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FRAMEWORK: OSeMOSYS / GLPK MathProg

HORIZON: 2021 – 2035 (15 Years)

REPO: github.com/kaustubh2005ashok-tech/EN401

STATUS: Published & Public

1. Executive Summary & Context

This explanatory document serves as the master technical blueprint and methodological guide for the coursework completed in **EN401: Energy Systems Modelling**. The project formulates and solves a national energy system capacity expansion and dispatch model using **OSeMOSYS** (Open Source Energy Modelling System) with the **GLPK** (GNU Linear Programming Kit) solver. Over the 15-year planning horizon (2021–2035), electricity demand expands from 1.05 PJ to 2.07 PJ. The primary challenge is balancing capital investments, operational expenditure, fuel costs, seasonal/diurnal renewable intermittency, and stringent carbon emission mitigation policies.

2. Repository Architecture & Version Control

The repository follows clean software engineering best practices, separating declarative problem models, computational inputs, raw solver outputs, and presentation deliverables:

```
EN401/
■■■ README.md # Repository landing page with quickstart
■■■ EXPLANATORY_GUIDE.md # Standalone technical architecture guide
■■■ EN401_Architecture_and_Methodology_Guide.pdf # This publication-quality PDF
■■■ .gitignore # Multi-tier exclusion rules (OS, lock, temp)
■■■ OSeMOSYS_Handouts_Report.pdf # 10-page executive analysis report
■■■ OSeMOSYS_Code_Explained.pdf # In-depth GNU MathProg code manual
■■■ OSeMOSYS_Results.xlsx # Master 10-sheet structured results workbook
■■■ H03/ # Handout 3 (OSeH03.dat, solution, PDF)
■■■ H04/ # Handout 4 (OSeH04.dat, solution, PDF)
■■■ H05/ # Handout 5 (OSeH05.dat, solution, PDF)
■■■ H06/ # Handout 6 (OSeH06.dat, solution, PDF)
■■■ Hands_on_1_to_6.docx # Original curriculum specification documents
■■■ extracted_images/ # 68 energy system diagrams & schematics
```

Version Control Design (.gitignore)

The repository includes a customized .gitignore designed for data science and modeling projects:

- **OS Metadata:** Excludes macOS .DS_Store, ._* , and Windows thumbnails to prevent filesystem pollution.
- **Office Locks:** Suppresses transient lock files (~\$*.docx, ~\$*.xlsx) created while workbooks or handouts are open.
- **Execution Noise:** Excludes scratch solver logs (*.log, *.tmp) and Python bytecode caches.

3. OSeMOSYS Mathematical Formulation

OSeMOSYS operates as a deterministic, bottom-up linear programming optimization framework. The primary decision variables determine annual technology additions (NewCapacity) and operational dispatch per timeslice (RateOfActivity) that minimize the total Net Present Value (NPV):

Objective Function:

$$\min \text{NPV} = \sum_{y \text{ in YEAR}} \text{TotalAnnualCost}_{\{y\}} / (1 + d)^{\{y - y_0\}}$$

Where TotalAnnualCost aggregates:

- + Capital Investment Cost: $\text{CapitalCost}_{\{y,t\}} * \text{NewCapacity}_{\{y,t\}}$
- + Fixed O&M Cost: $\text{FixedCost}_{\{y,t\}} * \text{TotalCapacityAnnual}_{\{y,t\}}$
- + Variable O&M Cost: $\text{VariableCost}_{\{y,t,m\}} * \text{AnnualActivity}_{\{y,t\}}$
- + Fuel & Import Costs: $\text{FuelCost}_{\{y,c\}} * \text{SupplyActivity}_{\{y,c\}}$
- + Carbon Penalty: $\text{EmissionPenalty}_{\{y,e\}} * \text{AnnualEmissions}_{\{y,e\}}$
- Salvage Value Credit: $\text{DiscountedRemainingLifeCapitalCredit}$

MATHEMATICAL CORE

Core Constraints:

1. **Demand Satisfaction:** Sum of grid output activity across modes $\geq$ Exogenous Demand for every timeslice.
2. **Capacity Availability:** Generation rate in timeslice $l \leq \text{TotalCapacity} * \text{CapacityFactor}(l) * \text{ConversionFactor}$.
3. **Lifetime Asset Tracking:** $\text{TotalCapacity}(y) = \text{ResidualCapacity}(y) + \text{sum of NewCapacity from surviving vintage cohorts}$.

4. Scenario Progression & Handout Analysis (HO1 – HO6)

The model is developed through six sequential stages, increasing in technological complexity, temporal granularity, and policy constraints:

Handout	System Scope & Key Changes	Timeslices	NPV System Cost	Dominant Technologies
HO1 & HO2	Formulation phase: Reference Energy System (RES) architecture, sets, and activity ratios	1 (Annual)	N/A	Conceptual mapping
HO3	Base electricity system: Coal (COA001), Hydro (HYD001), and Grid (TRN). Demand: 1.05 -> 2.07 PJ	1 (Annual)	\$25,321.4M	Coal generation baseload; Hydro restricted by high capital cost
HO4	Multi-fuel supply chain: Diesel imports (IMPDSL -> DSL001) and Gas mining (MINNGS -> NGS001)	1 (Annual)	\$25,321.4M (0.0% delta)	Coal solely chosen; Diesel and Gas unbuilt due to higher LCOE
HO5	Renewables & timeslice variability: 8 diurnal/seasonal slices; Solar PV (SOL001) & Wind (WND001)	8 slices (4 seasons x 2)	\$18,450.2M (-27.1% delta)	Solar PV dominates daytime; Hydro and Coal balance nighttime demand
HO6	Emissions constraints: CO2 emission factors by fuel, carbon tax, and annual emission caps	8 slices	\$21,140.8M (-16.5% vs Base)	Coal generation capped; rapid expansion of Solar PV, Wind, and Hydro

Key Methodological Insights:

- **Economic Least-Cost Merit Order (HO3 vs. HO4):** Introducing additional technologies does not alter the solution if their levelized generation cost exceeds the incumbent. Coal remains optimal until resource or environmental limits intervene.
- **Value of Temporal Granularity (HO5):** Slicing the year into 8 timeslices captures the coincidence between high daytime electricity demand and zero-marginal-cost solar generation, reducing total NPV cost by \$6.87 Billion (-27.1%).
- **Environmental Policy Trade-offs (HO6):** Imposing carbon penalties and caps raises NPV cost by \$2.69 Billion compared to unconstrained renewables (HO5), reflecting the shadow price of decarbonization.

5. Core Artifacts & Deliverables in Repository

- **OSeMOSYS_Results.xlsx:** Master 10-sheet interactive Excel workbook featuring Overview KPIs, scenario-by-scenario results, 15-year capacity addition matrix, cost comparison breakdowns, and a complete 17-parameter dictionary.
- **OSeMOSYS_Handouts_Report.pdf:** 10-page formal report outlining the analytical formulation, problem statements, technology deployments, and policy conclusions across all six handouts.
- **OSeMOSYS_Code_Explained.pdf:** Exhaustive software manual detailing GNU MathProg syntax, parameter indexing, equation mechanics, and GLPK solver flags.
- **HO3 to HO6 Model Directories:** Self-contained subdirectories containing the original MathProg .dat datasets, handout specifications (.pdf), and complete solver logs (_solution.txt).
- **Original Assignment DOCX & Images:** Unmodified source assignment handouts and 68 extracted Reference Energy System diagrams and technological curves.

6. Model Reproduction & Verification

All scenarios can be reproduced directly on any standard Unix or Windows terminal with GLPK installed:

```
# Solve any scenario using GLPK (glpsol):  
glpsol -m path/to/osemosys_fast.txt -d H05/0SeH05.dat -o H05/0SeH05_solution.txt  
  
# Inspect the objective value and primal solution:  
grep -E "Objective|TotalDiscountedCost" H05/0SeH05_solution.txt
```

VERIFICATION & DEPLOYMENT

Repository Status: Public on GitHub at <https://github.com/kaustubh2005ashok-tech/EN401>.

All code, data, spreadsheets, and reports are fully synchronized with origin/main.